

Part 3: Databricks Serverless Performance Modes

Part 2: [Databricks Warehouses](#)

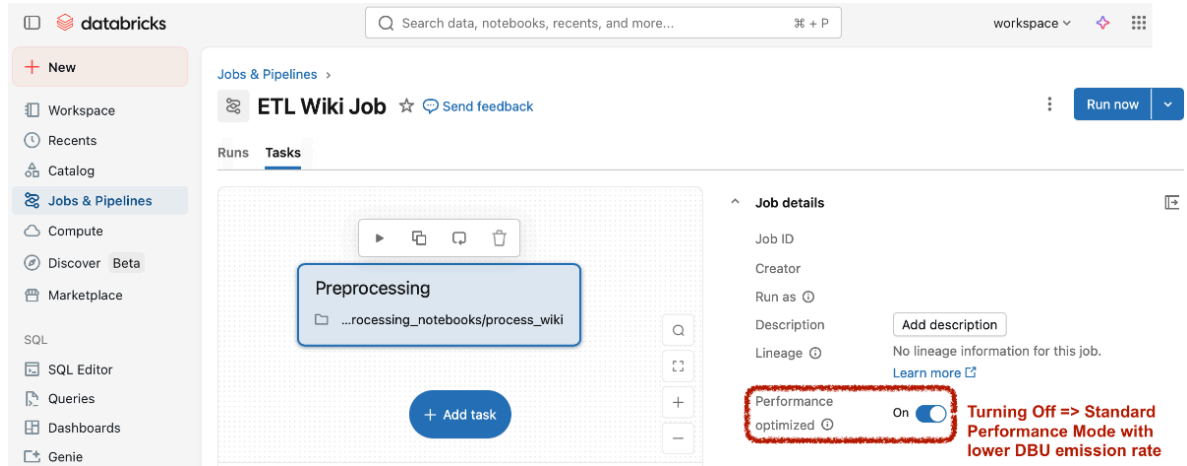
Two "performance modes" for Lakeflow **Jobs** with Serverless Compute exist:

- **Standard Mode**: Longer cluster bootstrapping phases, typically several minutes, similar to Traditional Compute
- **Performance Optimised**: Rapid launch times, instant compute, ideal for time-sensitive jobs

Both performance modes use JOBS_SERVERLESS_COMPUTE SKUs but the **emission rate** for the optimised setting tends to be significantly higher which likely results in **higher job costs**.

The optimised mode is activated by default upon serverless Job task creation, this is shown in the screenshot to the right.

The following [system table](#) query identifies the **most expensive** serverless job runs (using list prices) which were performance optimised and may become cheaper after switching to the standard mode:



```
WITH job_usage_opt AS (
  SELECT usage_metadata.job_name, usage_metadata.job_id, usage_metadata.job_run_id, identity_metadata.run_as,
  CONCAT(account_id, '#', workspace_id) workspace, * EXCEPT(account_id, workspace_id, usage_metadata, identity_metadata, product_features)
  FROM system.billing.usage
  WHERE usage_metadata.job_id IS NOT NULL AND usage_metadata.job_run_id IS NOT NULL AND product_features.performance_target = "PERFORMANCE_OPTIMIZED"
), job_usage_cost AS (
  SELECT * EXCEPT (prices.sku_name), job_usage_opt.usage_quantity * prices.pricing.default costs
  FROM job_usage_opt LEFT JOIN system.billing.list_prices prices ON job_usage_opt.sku_name = prices.sku_name
  AND prices.price_start_time <= job_usage_opt.usage_start_time AND (prices.price_end_time >= job_usage_opt.usage_start_time OR prices.price_end_time IS NULL)
)

SELECT job_name, job_id, job_run_id, run_as, SUM(costs) costs, currency_code currency, SUM(usage_quantity) DBUs, sku_name, workspace
FROM job_usage_cost
GROUP BY ALL ORDER BY costs DESC
```

Sample Query Results:

job_name	job_id	job_run_id	run_as	costs	currency	DBUs	sku_name	workspace
ETL_Ingestion	684766970	238850	myself@	6.51	USD	16.0	PREMIUM_JOBS_SERVERLESS_COMPUTE_EUROPE_IRELAND	123#456
ETL_Ingestion	684766970	315090	myself@	5.91	USD	15.1	PREMIUM_JOBS_SERVERLESS_COMPUTE_EUROPE_IRELAND	123#456
Benchmarking	594751973	929337	myself@	0.63	USD	1.61	PREMIUM_JOBS_SERVERLESS_COMPUTE_EUROPE_IRELAND	123#456
ML_Training	484066270	773466	myself@	0.62	USD	1.5	PREMIUM_JOBS_SERVERLESS_COMPUTE_EUROPE_IRELAND	123#456

More optimization queries can be found [here](#) made by  / [bigdatadude](#)